

ECON 1550

Spring 2026

Problem Set 9

Due: April 22, 2026 at 11:59pm ET

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Submission: Canvas or Gradescope

Instructions

- When submitting to Gradescope, indicate the page where each question is answered to avoid a 5-point deduction.
- Full credit is given for correct answers. If multiple steps are needed, you must show them to get full credit.
- Points are shown for each part. Partial credit is given for partially correct answers; show your work to maximize it.
- Late submissions receive a score of zero.
- If you have technical problems submitting, email your work to the Head TA before the deadline.
- Collaboration with classmates is encouraged; use of generative AI is permitted but discouraged.
- You must write, understand, and submit your solutions individually. Copying other students' or AI-generated answers, even fragments, is not allowed.

1. Multiple Choice (24 points)

Select the best answer for each of the following questions.

- (a) [4 points] Organizations whose members agree to fix their mutual exchange rates while allowing their currencies to fluctuate in value against the currencies of non-member countries are called
- ☐ (A) exchange rate unions.
 - ☐ (B) mutual rate unions.
 - ☐ (C) pegging rate unions.
 - ☐ (D) common rate unions.
- (b) [4 points] Which of the following correctly describes what happens when the central bank buys or sells assets?
- ☐ (A) Buying an asset increases central bank liabilities and expands the money supply; selling an asset decreases liabilities and shrinks the money supply.

- ☐ (B) Buying an asset decreases central bank liabilities and shrinks the money supply; selling an asset increases liabilities and expands the money supply.
 - ☐ (C) Buying an asset increases central bank liabilities and shrinks the money supply; selling an asset decreases liabilities and expands the money supply.
 - ☐ (D) Buying an asset decreases central bank liabilities and expands the money supply; selling an asset increases liabilities and shrinks the money supply.
- (c) [4 points] To hold the exchange rate constant,
- ☐ (A) a central bank must always be willing to trade currencies at the fixed exchange rate with the private actors in the foreign exchange market.
 - ☐ (B) a central bank cannot trade currencies at the fixed exchange rate with the private actors in the foreign exchange market.
 - ☐ (C) a central bank does not intervene to remove excess supplies or demands in the foreign exchange market.
 - ☐ (D) a central bank lets the foreign exchange and asset markets adjust on their own to equilibrium.
- (d) [4 points] By fixing the exchange rate, the central bank gives up its ability to
- ☐ (A) adjust taxes.
 - ☐ (B) influence the economy through fiscal policy.
 - ☐ (C) depreciate the domestic currency.
 - ☐ (D) influence the economy through monetary policy.
- (e) [4 points] Which one of the following statements is the most accurate?
- ☐ (A) Fiscal policy has the same effect on employment under fixed and flexible exchange rate regimes.
 - ☐ (B) Fiscal policy affects employment more under fixed than under flexible exchange rate regimes.
 - ☐ (C) Fiscal policy cannot affect employment under a fixed exchange rate but does affect output under a flexible exchange rate regime.
 - ☐ (D) Fiscal policy can affect employment under a fixed exchange rate regime, but does not affect output under a flexible exchange rate regime.
- (f) [4 points] Under fixed exchange rates, which one of the following statements is the most accurate?
- ☐ (A) Devaluation causes a reduction of the money supply.
 - ☐ (B) Devaluation has no effect on the stock of money.

- ☐ (C) Devaluation causes an expansion of the money supply.
- ☐ (D) Devaluation causes a reduction in output.

2. Short Question: Clean and Dirty Floats (6 points)

What is the difference between clean and dirty float exchange rates?

3. Fixed Exchange Rates (10 points)

- (a) [5 points] Explain one advantage and one disadvantage of fixed exchange rate regimes compared to having a fully flexible exchange rate.
- (b) [5 points] Assume the economy is at its long-run equilibrium and that the current account is in balance (equal to zero). The central bank is implementing a fully credible fixed exchange rate regime. Using the *AA-DD* model, explain why a devaluation improves the current account.

4. Flat AA and Fixed Exchange Rates (10 points)

Consider the following version of the AA-DD model:

Consumption	:	$C = (1 - a)Y$
Investment	:	I
Government spending	:	G
Current account	:	$CA = b\frac{EP^*}{P} - cY$
Aggregate demand for goods and services	:	$Z = C + I + G + CA$
Money demand	:	$\frac{M}{P} = 1 + d(Y - Y^f) - eR$
Money supply	:	$\frac{M}{P} = \frac{\bar{M}}{P}$
Uncovered interest parity	:	$R = R^* + \left(\frac{E^e}{E} - 1\right)$

The variables P^* , I , G , Y^f , $\bar{M}$, and R^* are exogenous. The parameters a , b , c , d , and e are also exogenous and assumed to be positive. The variables C , Y , CA , E , P , Z , E^e , M , and R are endogenous.

The price level P is fixed in the short run but fully flexible in the long run. Assume that, initially, the economy is at its long-run equilibrium with output equal to the full-employment level Y^f , where Y^f is an exogenous variable.

The above model is the standard “textbook” version of the AA-DD model we have been using in class.

- (a) [5 points] For this question, assume that the economy starts at the long-run equilibrium that is reached after d changes to zero permanently, and continue to assume that $d = 0$ permanently. Consider a temporary fiscal expansion that increases G to a higher value G^{high} . In the short run, does the current account improve (increase), deteriorate (decrease), or stay the same? Give economic intuition for your answer.

- (b) [5 points] Now assume that, before the temporary fiscal expansion, the central bank implements a fixed exchange rate regime at the prevailing exchange rate (which is the long-run level with $d = 0$). Are the short-run effects of fiscal expansion on the current account the same as in (a)? Why or why not?

5. A Preview of the II-XX Model (50 points)

In this problem, we want to chart the state of the US economy in a plot that has Y on the horizontal axis and q on the vertical axis, as in the *AA-DD* model but with the real exchange rate q rather than the nominal exchange rate E on the vertical axis. The goal is to have a look at what the data looks like so that when we learn the II-XX model we can have a sense of how applicable it is, and what the data patterns are like. We will use data from the St. Louis Fed's [FRED](#).

We use the series NGDPPOT as our measure for the full-employment level of output Y^f . We use the series GDP for output Y . Last, we use the series RBUSBIS as a measure of the real exchange rate q . Download annual frequency data¹ starting in 1994 (the first year for which the exchange rate data is available) and ending in the most recent period that is available for all data series.

- (a) [10 points] Briefly explain how the series NGDPPOT is constructed.

Hint: Follow the link from FRED to the Congressional Budget Office and search for the document that explains how potential output is constructed.

- (b) [10 points] Explain what RBUSBIS measures.

- (c) [10 points] Construct the following measures:

- Output gap = $GDP/NGDPPOT - 1$
- Normalized output gap = $\frac{\text{Output gap} - \text{mean of output gap}}{\text{standard deviation of output gap}}$
- Normalized exchange rate = $\frac{RBUSBIS - \text{mean of RBUSBIS}}{\text{standard deviation of RBUSBIS}}$

Make two scatterplots with the normalized output gap on the horizontal axis and the normalized exchange rate on the vertical axis. The first scatterplot is for the years 1994-2007, and the second one for 2008 to the end of the sample. Try to see if you can get the plot to show lines connecting the data points, e.g., the 1994 data point connects to the 1995 data point with a line segment, the 1995 data point connects to the 1996 data point with another line segment, etc.

- (d) [10 points] The *II-XX* model makes specific predictions regarding what the two scatterplots should look like. In some situations, the model predicts that the economy

¹If the original data comes at quarterly frequency, FRED will convert it for you before you download it, so you don't have to do the work. You can choose how you want the conversion to be done: you can choose to average across quarters, you can pick the value in the last quarter, etc. Any of the options are OK, pick the one you think makes the most sense to you.

should evolve following a clockwise spiral. In other situations, it predicts a counterclockwise spiral. Do your two scatter plots look like they follow a spiral? If yes, is it clockwise or counterclockwise? If not, can you come up with a simple way (i.e., in a few words) to describe the evolution of the economy?

- (e) [10 points] Now make a single scatter plot with all the years. Looking at this new scatter plot, do you see any new patterns? How does this scatterplot with all the years compare to the other two scatterplots from (c)?